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Dating apps: a literature review

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ABSTRACT

Dating apps have become one of the most prominent and contentious topics in the realm of intimacy among the wider public and academia. Media and communication researchers have examined their uptake across cultural contexts, seeking to address the dynamics between dating apps and social processes. With the knowledge accumulated in this research field, we assemble a comprehensive account of interactions through dating apps. We categorize existing findings about dating apps into three sections: dating apps and their reconfiguration, dating practices and their remediation, and lastly social arrangements and their reformation. These sections together present dating apps as a technological consequence of various social forces that mediate users' daily practices and social relationships.

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Introduction

In the last few years, mobile dating applications, or 'dating apps,' have drawn lots of attention from researchers who examine the dynamic between new media technologies and society. As location-based services, dating apps run on smartphones and work with the global positioning system (GPS). Affording both synchronous and asynchronous communication, they connect users in geographic proximity; many also allow users to browse the profiles of those who are half a world away. Unlike traditional dating websites, dating apps significantly shorten the period between the initial online contact and subsequent offline encounters. Unlike mainstream social networking platforms such as Facebook and WeChat, dating apps operate on users' expectations of connecting with strangers.

Regarding their design features, dating apps can effectively be categorized into two types. One type allows the user to start a conversation through private messaging with any user displayed on the screen. Apps of this type often have a grid view or a list view, presenting a range of nearby users' profiles in descending order of geographic proximity. This type includes the most popular queer-targeted apps, such as Blued and Grindr. The other type entails a mechanism of signalling and matching, as private messaging is possible only when both users signal their interest. A prominent representative of this type is Tinder, whose users are mostly heterosexuals. Tinder presents one single profile at a time. Users need to swipe left or right on the profile to signal their dis/interest in establishing a connection. Lastly, a dating app can blend the two aforementioned modes.

Dating apps can hardly be defined as 'apps for dating' if 'dating' means the social meeting that is aimed at developing a romantic relationship. 'Match. Chat. Date.' That is the slogan Tinder displays on its official website (Tinder, 2020). However, there is a much wider range of user motives in reality, including casual sex and simply killing time (Timmermans & De Caluwé, 2017a). Unlike Tinder, Grindr

sells a sense of community and brands itself as a 'social networking app for gay, bi, trans, and queer people' instead of a dating app (Grindr, 2020). Meanwhile, many researchers refer to Grindr as a 'hook-up app' (e.g. Davis et al., 2016; MacKee, 2016; Race, 2015a), which downplays its role in romantic relationship development. Blued, the Chinese counterpart of the Western Grindr, has even integrated the functionality of live streaming along its evolutionary path, which sets it apart from the stereotypical image of a dating app (Wang, 2020).

The landscape of dating apps is more diverse than the word 'dating' suggests. It warrants scrutiny. What are the technological features that make dating apps new? What are the factors that have shaped these technological features? What do people actually use dating apps for? How do dating apps mediate users' daily practices and social relationships? These questions appeal to researchers who take various approaches to understand the interplay between dating apps and our society.

In this article, we review dating app studies and piece together the answers researchers have got regarding the above-mentioned questions. To do so, we adopt the mediation framework proposed by Lievrouw (2014), which is a good fit for understanding the mutual shaping between communication technologies and society. This framework suggests we grasp three components: (a) *artefacts*, devices or objects with certain technological and material features, used by people to communicate with each other; (b) *practices*, how people engage in communication with devices; and (c) *social arrangements*, social relations, institutions, and structures that not only organize and govern, but also form and develop around communication technologies and practices. These three components are in a constant state of flux. There are three corresponding modes of change: the reconfiguration of artefacts, remediation of practices, and reformation of social arrangements. Lievrouw (2014, p. 45) elaborates on this:

Artefacts – material devices and objects – enable, extend, or constrain people's abilities to communicate, and develop through a process of reconfiguration. People engage in communicative practices or action, some of which may employ those devices; practices change in an ongoing process of remediation of interaction, expression, and cultural works. Social arrangements – patterns of relations, organizing, and institutional structure – form and develop in concert with the artefacts and practices through a process of reformation.

Address artifacts, practices, and social arrangements simultaneously, this tripartite framework affords a more nuanced analysis than other social constructivist approaches, including the 'social shaping of technology' (SST) approach that pays less attention to users' appropriation of technologies (Mackay & Gillespie, 1992), and the domestication approach that hardly examines the production of artifacts (see Berker et al., 2006).

Wu and Ward (2018) apply the mediation framework in an earlier review article, on gay dating app studies. They frame dating apps as artefacts and understand the ways people use the apps as practices. The social relations implicated in the configuration and, especially, the use of dating apps are social arrangements. Wu and Ward (2018) thus present researchers' findings in two sections: the remediation of gay men's dating practices and the reformation of social arrangements around gay life. They also point out the lack of research on the reconfiguration of gay dating apps.

In this review, we apply the mediation framework in the same way as Wu and Ward (2018). Nevertheless, we have a much larger scope, not solely examining gay dating app studies. Besides, we find that studies on the reconfiguration of dating apps are fruitful, due to our larger scope and the timing of this review. Accordingly, we present the findings in existing dating app studies in three sections: (a) dating apps and their reconfiguration; (b) user practices and their remediation; (c) social arrangements and their reformation. Although the word 'reformation' often has a positive connotation, we use it in a neutral sense, as we avoid seeing changes in social arrangements as essentially positive or negative. Besides, even those commonly acknowledged positive changes will not be spread evenly among individuals. Instead of offering our judgment, we aim to explicate researchers' opinions and the reasoning behind them. Note that some researchers are interested in how some social

arrangements are reproduced instead of being reformed. We put their findings in the third section as well, as a contrast to dating apps' transformative role that is discussed by other researchers.

Methods

Before we started writing this review, the first author had been tracking dating app studies for four years. He compiled a list of the studies he sought or randomly encountered. To supplement the list, a systematic collection of relevant studies was conducted on the Web of Science, a leading science search platform that is often used for literature analysis and allow users to access multiple interdisciplinary databases (Kubin & von Sikorski, 2021). We limited the database to the Web of Science Core Collection so that the quality of the studies we find would be high. When choosing the research areas, we focused on media and communication studies, as well as those we deem closely related to this area. Therefore, in the system, we selected 'communication,' 'behavioural sciences,' 'psychology,' 'social science other topics,' and 'sociology.' We also limited document types to 'articles,' and languages to 'English.'

We performed an initial search on 29 November 2021, using the terms 'dating app,' 'hookup app,' 'hook-up app' and their plural forms. This yielded 260 results, among which the most often mentioned dating apps were Tinder and Grindr. Considering that some researchers may not refer to these two apps as dating apps or hook-up apps, we did another two rounds of search on 1 December 2021, using the names of the two apps respectively as the topic. Notably, the word 'tinder' had long existed in English. Thus when Tinder was the search term, we only looked at publications since 2012, as Tinder was founded in that year. We also filtered out the results that contained any of the six search terms we used in the first round. This led to 75 results for Tinder and 77 for Grindr. Among all the 412 results, many were centred on public health, especially on HIV prevention. We do not engage with these studies, given that our target readers are media and communication researchers. In this review, we only present studies that well elucidate the dynamics between artifacts, practices, and social arrangements related to dating apps.

Dating apps and their reconfiguration

Technological artefacts are shaped by social forces. Dating apps are no exception. Technological features of dating apps have drawn attention from researchers who aim to understand these features as a result of social shaping. For instance, Tinder used to require its users to log in with their Facebook accounts. Duguay (2017) explains this as a solution to users' potential concern about authenticity. Before the rise of dating apps, online daters' most prominent concerns had been that others were misrepresenting themselves and that their safety might be threatened when meeting strangers in offline settings. Tinder needed to convince its users that the identification information of their potential dates is authentic. Connecting via Facebook means that the user's information such as his or her first name, age, recent photos, and gender is imported into the Tinder profile from his or her Facebook profile. This is seen as a guarantee of authenticity because Facebook has promoted a real name culture since its foundation by policing the use of real names and suspending accounts suspected of using fake names.

A few years after Duguay published her study, Tinder started to allow users to sign up with their phone numbers, and a Facebook account was no longer necessary. Indeed, dating apps never stay the same. They are in a continuous process of reconfiguration, accompanied by design iteration and rebranding. One of the earliest studies on the reconfiguration of dating apps was carried out by Murray and Ankerson (2016). In their case study of the lesbian dating app Dattch, first launched in 2013, Murray and Ankerson reveal how the app has been shaped by the interplay between iterative design, technology start-up culture, and perceptions of lesbian sexuality. The earliest versions of Dattch foregrounded the location-based model of Grindr, with the main interface presenting a grid of nearby users' profiles. This seems unsurprising, given that the first dating apps aimed at lesbian

users were mostly 'reskins' of popular gay apps. The practice of 'app reskinning' is common in the internet industry: Developers take an existing app as a starting point, change its design and artwork, and rename it, hoping to replicate its success with an imitation. Nevertheless, Dattch's user activity data showed that proximity-led contact did not work well with lesbian users. To facilitate interactions between users and thus keep them coming back to the app, design changes were made at different points to facilitate conversations and thus a sense of connection, even though physical contact may still stay at a slower tempo. For instance, the newly designed interface invited users to share their lives by posting selfies, food pictures, and holiday photos and inputting texts about themselves. These types of input are supposed to serve as prompts for conversations. Overall, the efforts in retaining users and enlarging the user base largely result from the technology start-up culture and were backed by financial investments. After all, start-up companies need to prove to their investors that their products are scalable and have real potential for mass popularity. In 2015, the company rebranded Dattch as Her and expected it to attract female users with divergent social-sexual positions on the continuum of lesbian identification.

The reconfiguration of dating apps has received lots of attention from Chinese researchers. A common theme in their studies is the significant role played by the state in the shaping of Chinese dating apps. Liu (2016) presents the case of Momo, the first location-based Chinese dating app. At an early stage of its development, Momo was widely used for casual sex and thus had gained the reputation as 'a genius tool for getting laid.' The erotic photos and pornographic content circulating on Momo among its users drew attention from the National Sweep Away Pornography and Strike Down Illegal Publications Task Force, a coordinating body involving various agencies that oversees the Chinese public sphere. The company received an administrative penalty and a warning. To secure itself, Momo launched a 'sanitizing' project to 'desexualize' itself. For instance, the main interface used to present the profiles of nearby users in a list view, which was perceived to facilitate casual sex between strangers. Then it was replaced by an interface with social feeds of users, comparable to the Facebook wall or WeChat Moments. The functionality of searching for nearby users remained, but it could only be found in a secondary menu. Other 'sanitizing' measures include removing chat groups from the app that were associated with controversial sexual activities such as partner swapping, group sex, one-night stands, and sadomasochism, as well as banning the accounts that were reported to be distributing 'unhealthy' content. Momo also carried out a publicity campaign to inform the public that it is an app for a wide range of social activities, but not for casual sex.

The Chinese dating app that received the most scholarly attention is Blued, a gay-targeted app. Miao and Chan (2020) have examined how the state, together with the capital market and visions of the company founder and staff members, shapes the developmental trajectory of Blued. Initially built as a reskin of the Western gay dating app Jack'd, Blued was meant to cater to the sexual needs of its users. To survive in a country where LGBTQs are less visible in the public sphere and less tolerated by society, Blued chose to collaborate with the state closely. Specifically, it has actively participated in the HIV prevention work led by the state, positioning itself as a 'health education platform.' For instance, information about HIV prevention is disseminated to its users through private messages and public advertisements on the app; the company also provides HIV tests at its headquarter in Beijing. Due to the political pressure, however, Blued carefully keeps a distance from LGBTQ rights organizations. As the state makes lots of efforts to crack down on pornographic online content, Blued has a content management team responsible for reducing the pornographic content on the app.

As Miao and Chan (2020) show, Blued has gradually deviated from the company's original vision for a gay hook-up app and repositioned itself as a 'social app.' The repositioning not only serves the maintenance of a positive relationship with the government, but also helps Blued to win the favour of investors. Driven by the logic of capital, an internet company like Blued tends to 'accumulate as many users as possible, retain them, and then monetize them' (Miao & Chan, 2020, p. 220), which also can be seen in Murray and Ankerson's (2016) study on Dattch/Her. The challenge facing Blued is to

enhance its social affordances and increase users' attachment to the app. Therefore, design features have been revised. For instance, Blued added the 'Post' function in 2013, allowing users to post 'statuses' as on mainstream social networking sites. It also added the 'Groups' function in 2014, allowing users to set up discussion groups for hobbies and interests. The most drastic change, however, was to integrate the functionality of living streaming into the app structure. Live streaming is now the most profitable division of Blued's core business.

Indeed, the reconfiguration process of Blued is quite different from that of its Western counterparts. Blued has integrated many new functionalities, such as gaming and online shopping. As Wang (2020, p. 518) has noted,

Although Blued itself is not a game developer, a number of game products embedded in the app require users to log in via Blued accounts. Likewise, its online shop is built with software development kits from other technological service providers.

On the other hand, content posted on Blued can also be shared with mainstream social media platforms, such as WeChat and Weibo. Accordingly, Wang (2020, p. 504) argues that Blued is undergoing a process of 'platformization,' understood as 'a dual process of outward extension into other websites, platforms, and apps and inward integration of third-party plug-ins and tabs.'

User practices and their remediation

Uses and motives

When it comes to speculating how people make use of dating apps, public discourse such as press coverage often relies on simplistic accounts, collapsing various motives for using apps into a single category of practice. Yet we can speculate that users are diverse and that empirical accounts of prominent apps would suggest the pursuit of a range of romantic affiliations, from casual sex to more enduring companionship, as well as practices fully detached from erotic or romantic pursuits. Users may turn to these sites to seek other kinds of information, affiliation, and security, especially sexual minorities and members of marginal communities who lack other opportunities to engage with like-minded people. Uses and motives also vary based on demographics as well as personality traits. Empirical work in this area, notably survey data of users, supports a diverse set of motivations and intended uses for dating apps.

Sumter et al. (2017) go beyond the so-called 'hook-up culture' to explain the appeal of dating apps, seeking to explain the plurality of motivations why emerging adults use Tinder. Drawing upon Uses and Gratification Theory (UGT), they account for a range of motivations, but also how these may differ on the basis of age, gender, frequency of use and offline outcomes. This design seeks to yield in-depth knowledge of prominent social platforms, which the authors in turn believe can help prevent apparent negative outcomes. The authors conduct an online survey, based on a convenience sample of Dutch 18–30-year-olds. They identify six primary motivations that include sexual, communicative and self-validating intentions, and that also differ according to age and gender. One overarching finding is that users are on the whole more motivated to find love on dating platforms, as opposed to casual sex. This suggests that Tinder and its less prominent counterparts should be understood in public and academic discourses as being sought for more than either fleeting hook-ups or conventional dating.

Other scholars also make use of UGT to expand scholarly understandings of the range of motives that might compel someone to make use of mobile dating apps like Tinder. Timmermans and De Caluwé (2017a) developed a scale of motivations based on a series of qualitative and quantitative studies in the Belgian context. They arrived at a set of thirteen measures, which expand the oft-repeated focus on casual sex in order to include more social measures such as social approval and peer pressure, as well as measures associated with personal fulfilment such as curiosity, belongingness and entertainment. Based on the prominence of these motives, they propose that it is more appropriate to consider Tinder as a 'location-based screening/meeting application' rather than a

'mobile dating application' (Timmermans & De Caluwé, 2017a, p. 348). They also acknowledge that the prominence of these uses may vary per country, and that frequency of use is surely also an important factor in further refining our understanding of what brings users to these applications.

Miller (2015a) also accounts for sought gratifications in exploring motivations for social networking use among men who have sex with men (MSM). Also drawing upon UGT as conceptual grounding, he positions his study in relation to the increased popularity of online queer spaces and MSM social networking as 'nothing short than revolutionary for the LGBTQ community' (Miller, 2015a, p. 481). In this context, the 'virtual' matters particularly more for dating, and they are explicitly framed as social networks due to the possibility of forging bonds and sustaining a community. Drawing on an online survey of current or former users, Miller identifies seven non-mutually exclusive motivations. A cross-cutting theme with these motivations is being able to do what is otherwise not possible in the offline world (or rather, in a world outside of these apps, including bars and clubs in the case of gay men). They also stand in comparison to older online technologies, including regional chat rooms. We can arrange motivations into three clusters:

First, safety and control. Safety refers to the assurance that everyone on the app can be assumed to be gay. Not only is there minimized risk of endangerment to all involved participants, but this also ensures a kind of emotional safety. Control refers to the agency over who they interact with, how much they disclose. This includes the ability to screen and block unwanted advances, as well as not having to respond to such engagements. This allows users to assess a potential partner's ethics and values in a way that saves time, and can also streamline sociality.

Second, MSM dating apps facilitate and enable practices. This includes a drive towards connectivity, and Miller notes that some respondents feel deeply about 'being a part of something larger' to establish a positive queer identity (Miller, 2015a, p. 481). Yet respondents also state that this connection can be superficial, suggesting a potential disparity between gratifications sought and received (Miller, 2015a). A related motivation is the ease with which apps allow users to meet men, compared to offline environments. This is related to perceived transparency of intentions, partly how the app is framed, but also a reflection of those who use it, based on which boxes they check. Respondents also turn to these sites due to their accessibility, in which male sexual minorities can make themselves available to each other, notably for those who are uncomfortable with 'scene'-based alternatives like bars and clubs.

Third, mobility and versatility allow diverse and serendipitous encounters. Here mobility refers to geolocate features that allow users to connect with others nearby, though proximity is not always what is sought. This is useful when visiting a new city, but also when planning to meet others in a specific location, such as a future travel destination. The versatility motivation suggests that users are not just interested in sex, a finding that also emerges in China's survey of MSM-based dating app users (Chan, 2017a). Respondents complain about other users too often seeking to hook up, even though they also do value this possibility. While his findings suggest a range of motivations linked to diverse forms of user agency, Miller notes the need for future work on the realization of gratifications, given the disjointedness between gratifications sought and obtained.

The above studies explore a range of uses that fit within the expectations of app designers. Other practices may be unanticipated by designers and owners, who may intervene in response. Duguay (2020) draws upon Science and Technology Studies (STS) to provide an empirical account of 'off-label' and thus unanticipated practices, as well as subsequent (re-)appropriation in the case of Tinder. The term off-label invokes a useful pharmacological imagery, though the author also acknowledges the limits of this frame when it comes to matters like regulation. Her study aims to define and situate off-label uses, which facilitates an understanding of relations between platforms and users. It charts the uptake and implementation of a platform, in which a broader set of stakeholders develop normative expectations, that not all users may carry. The author identifies four kinds of use, in order to provide an analytic lens: (1) on-label conventional use, (2) expected use, where users distinguish how an app *should* be used from how it can be used, (3) off-label use that contravenes or circumvents on-label use, and (4) platform response, which may adopt some

novel dimensions of off-label practices. Duguay takes a mixed-methods approach, combining a 'walkthrough' (Light et al., 2018) account of platforms with analysis of press coverage of controversial Tinder uses, and interviews with instigators of off-label uses. This suggests that off-label uses are legally and morally dispersed (such as anti-smoking campaigns vs drug dealing), and thus do not constitute a coherent form of platform deviance. They often build on established expected use (i.e. expectations of hooking up) as a means to make contact with other users. Other studies note that off-label uses may provide much-needed opportunities for marginalized individuals, as when non-Western immigrants in Copenhagen use Grinder to find jobs and accommodation (Shield, 2019). It bears noting that such practices can also be a source of risk for users facing vulnerabilities. Duguay reports that women received harsher backlash upon discovery of off-label users, for example through sexist vitriol. Reflecting on the platform's response to off-label use, the author suggests more recent Tinder-supported ads resemble off-label initiatives, while the direct influence of marginal practices on the mainstream of the platform is difficult to establish.

Recent studies also consider demographic and dispositional factors in relation to different uses and motives. When focusing on both the intent to seek romance and casual sex on dating apps, it is possible to consider personality characteristics as well as technology-based behaviour as determining factors. Chan (2017b) examines relationships between trust towards people online, sensation seeking, smartphone use for accessing the internet and the intent to use dating apps to look for romance and casual sex. He draws upon past research on online dating, in which personal attributes like trust are understood as strong predictors of using dating websites. While one may expect similar findings when shifting from websites to apps, this transition is significant in terms of socio-technical context. Chan conducts a survey of heterosexual men and women in order to identify three personal attributes: (1) trust towards people online, (2) sensation seeking, and (3) smartphone use for accessing the internet. Taking a path model approach, his results suggest a direct link between both sensation seeking and smartphone use with behavioural intent, while 'the relationship between trust and behavioural intent was fully mediated by a belief variable' (p. 256). His findings also suggest that perceived norms played a role in looking for casual sex via dating apps, while 'attitudes were found to be a significant predictor of the corresponding intent' (p. 253) for the pursuit of romance and sex.

Likewise, Ranzini and Lutz (2017) also consider how user motivations make deliberate use of the perceived affordances of mobile apps like Tinder. They consider both demographic and psychological factors, including motivational and psychological antecedents such as self-esteem, loneliness and narcissism. The authors obtained survey data to determine how Tinder users present themselves. They find self-esteem and motive or purpose of Tinder use to be the strongest predictors of self-presentation. Likewise, high self-esteem is linked to less deceptive and more authentic forms of self-presentation, while self-validation had the strongest effect on self-presentation. Gender differences in uses are prominent, with men primarily pursuing hook-up sex, travelling and relationships, while women are more inclined to seek friendship and self-validation. These findings align with earlier literature on social media use and gender. In terms of sexual orientation, heterosexuals are said to present themselves in a more 'authentic' manner than homosexuals, due to the imposition of heteronormativity onto such practices and platforms.

Timmermans and De Caluwé (2017b) also contribute to relations between personality traits and mobile dating application uses and motives, building on their earlier work in this area. They apply the big five personality traits to single Tinder users as well as single people who have not used the app. Their findings suggest that those who rate high on agreeableness are less likely to be motivated by the pursuit of sexual experiences, while high conscientiousness respondents are more driven to seek a relationship, rather than treat the app as a source of entertainment or distraction. This approach seems to resonate with other studies that understand personality traits shape motivations to turn to dating apps, which as a result may shape the emergence of antisocial and other problematic behaviour (Orosz et al., 2018). The authors note that these findings help describe a contemporary understanding of Tinder as both an extension of conventional and embodied romantic activities,

as well as a significant shift from such activities, especially for those who may not feel comfortable first pursuing romance in a face-to-face setting. Their subsequent work with Alexopoulos not only distinguishes single and non-single Tinder users, but also considers maladaptive personality traits such as the 'dark triad' (Timmermans et al., 2018). Their online survey finds that Tinder users in a relationship are less driven by motives such as relationship seeking and flirting, and more compelled by 'their own curiosity about the current dating market' as well as 'their own value as a potential dating partner' (Timmermans et al., 2018, p. 135). This suggests a distinction between using Tinder for immediate romantic pursuits, and prominent uses and motives that are framed more in terms of hypothetical or potential romance. Other survey data present dark triad personality traits as related to short-term mating motivations (Sevi, 2019), while light triad traits (Faith in Humanity, Humanism and Kantianism) are associated with higher motivation towards long-term mating (Sevi & Doğruyol, 2020).

Some scholarship also focuses on emerging apps for establishing romantic relations, including with non-users. This enriches our perspective beyond why some make use of these services, in order to also ask why others end up not using these platforms, despite a potential desire to do so. Sumter and Vandenbosch (2019) employ the Media Practice Model, conducting a survey of university students as well as the broader population in the Dutch context. Like others, they consider the link between dating app uses and motivations, and demographic as well as personality-based variables. In terms of factors linked with being more inclined to use dating apps, their results point to variables such as non-heterosexual identity, low dating anxiety, and more sex permissive attitudes. In addition, ease of communication is a relevant motivation for people with high dating anxiety, as well as men in particular. We may speculate that other mobile dating apps targeting a niche market may position themselves as suitable for those who do not fit this profile (such as Bumble), though it remains unclear whether such apps reach different user groups in terms of personality, or if these patterns are generalizable across all available applications. Anxiety as it relates to attachment styles seems to play a pivotal role in shaping dating app motives. Chin et al.'s survey findings (2019) suggest that dating app users with anxious attachment were motivated by a desire to meet others, in comparison to those with an avoidant attachment style. In the case of MSM users in particular, Jayawardena et al. (2021) claim that the former sought self-esteem enhancement in Grindr use, whereas the latter were motivated by escapism in their usage.

Researchers also point out that users' specific goals are negotiable. In a study conducted in the United States, Fitzpatrick and Birnholtz (2018) explore how Grindr users negotiate their goals in different stages of interaction. The initial negotiation starts with constructing one's own profile. Given that goals are negotiable and may vary in different situations, stating a specific goal in the profile makes it difficult to withdraw this information without undermining self-consistency. In contrast, retaining some ambiguity in the profile can leave some room for later negotiation in the interaction. When it comes to the private chat stage, the timing of the other user's reply, whether immediate or delayed, can impact the flow of the chat and also one's previous expectations. Finally, face-to-face meetings arranged through dating apps are another stage of negotiation, as users either confirm or overturn the previous online impression they had of their dating partners. Grindr users thus build, shift, and negotiate their anticipations in different interaction stages (Bonner-Thompson, 2021).

Self-presentations in profiles

Self-presentation, or impression management, is a prominent theme in dating app studies. As Ward argues (2017, p. 1645), 'dating apps present a novel technological environment for impression management, mainly due to issues of reduced cues and increased control, local proximity, and a reduced filtering process.' However, the concern for the authenticity of self-presentation in a mediated environment is not new (Baym, 2015). Unsurprisingly, it has been repeated in dating app studies. In one of the earliest studies on Tinder, Ward (2017) found in the Netherlands that users strike a

balance between an ideal and an authentic self-presentation. In line with past research on traditional dating websites (Ellison et al., 2006), this finding suggests that authenticity is still valued by online daters. Nevertheless, the degrees of authenticity vary with users' demographic and psychological factors. This is discussed in the above-mentioned study by Ranzini and Lutz (2017).

Apart from authenticity, masculinity is also a keyword, particularly in MSM-specific studies. Interested in MSM's self-presentations and descriptions of partner preferences, Miller (2015b) conducted a content analysis of three hundred profiles worldwide from the MSM-specific dating app Jack'd. Overall, the findings suggest that Jack'd users tend to privilege masculinity, present themselves as topless in profile pictures, and mention fitness or bodies in profile texts. Although masculinity can be presented visually through profile pictures, some men still present themselves as masculine linguistically. Compared to users who have shown their faces in profile pictures, those who have not are more likely to mention fitness or bodies in self-descriptions. Miller (2015b) has also noted some racial differences in this study. Black users are more likely to make self-descriptions about their masculinity, use faceless profile photos, and discuss their sexual positions. By contrast, Asian users are less likely to disclose their sexual position preferences. Regarding the descriptions of partner preferences, it is notable that racial preferences are not frequently mentioned. However, the sample of this study is limited in the sense that the users' geographic locations are unknown. Therefore, the above-mentioned findings are dissociated from specific cultural or national contexts. They underscore the importance of cultural context, as different societies may not share the same ideal of masculinities. Chan (2021b), for instance, refers to the traditional Chinese masculine ideals of *wen* (cultural attainment) and *wu* (physical prowess) when examining the performance of masculinity on dating apps in China. The *wen-wu* dyad suggests a dualistic understanding of masculinity that does not prevail in the Western context.

In another three studies on MSM's self-presentations in dating app profiles, Miller is more confident to contextualize the findings, with most of the participants from the United States and Canada (Miller, 2018, 2019; Miller & Behm-Morawitz, 2016). Miller and Behm-Morawitz (2016) conducted an online experiment to test how MSM respond to anti-effeminate, femmephobic language use in dating app profiles. Specifically, they wanted to know whether a man's using femmephobic language impacts (a) the way he is perceived and (b) how willing other men are to interact with him. Regarding the first focal point, results show that the use of femmephobic language does not significantly change others' perceptions of a man's physical attractiveness, probably because femmephobic remarks reflect one's personality rather than physicality. Nor does it significantly change how others perceive a man's masculinity, which means a man would not be perceived to be more masculine simply because he uses femmephobic language. However, perceptions of other aspects can be affected: Men with femmephobic profiles are perceived to be less intelligent, less sexually confident, and less desirable as romantic partners. As for the second focal point, users are less willing to interact for friendships or romantic purposes with a man when he uses femmephobic language than when he does not use such language. Nevertheless, femmephobic language does not seem to influence how willing MSM are to have sex with someone.

Miller (2018) explored three types of languages that are related to the phenomenon of masculinity privileging on MSM-specific dating apps, namely masculinity language (e.g. 'Masc guy here,' 'No fems,' etc.), body language (e.g. 'Looking for muscular and beefy guys,' 'I don't have a six-pack,' etc.), and sports language (about sports and fitness/working out). Among these three languages, sports language is the most prevalent one, while masculinity language is the least. Regarding body language, there is a mix between men discussing their own or others' muscularity and men discussing their lack of muscularity. On average, men who use body language are older than those who do not. The average education level of men who use sports language is higher than that of those who do not.

Miller (2019) also examined face-disclosure and body-disclosure in MSM's visual self-presentation on dating apps. In his findings, 'face-disclosing images were connected to higher levels of app usage, longer-term app usage, and degree of outness, while having body-disclosing photo was related to

age, drive for muscularity, self-perceived masculinity, and level of anti-effeminacy' (Miller, 2019, p. 15). In other words, men who are more likely to show their faces in profile photos are those who use dating apps more frequently, who have used apps for a longer term, or who are more open about their queer identities. Meanwhile, men who tend to have shirtless photos in their profiles are those who perceive themselves to be masculine or who hold stronger anti-effeminacy attitudes. Their average age is older than that of other users.

Bonner-Thompson (2017) argues that there are two forms of masculinities produced through Grindr profiles, namely hypersexual masculinity and lifestyle masculinity. Hypersexual masculinity is produced through body-centred photos. In these photos, the body takes up most space, and the background is obscured in these photos. The absence of a recognizable context gives rise to the hypersexualized embodiment. In comparison, lifestyle masculinity is produced through pictures in which the body is situated in a particular context, such as a beach, a bar, or a music event. It emerges when Grindr users 'seek to (re)construct a digital body that is entangled with everyday and material geographies' (Bonner-Thompson, 2017, p. 1619). Not being mutually exclusive, hypersexualized and lifestyle masculinities can intersect in Grindr profiles.

Notably, the correlation between face-disclosure, body-disclosure, and the degree of outness is a recurring theme in queer men specific studies (Lemke & Weber, 2017; McGuire, 2018; Miao & Chan, 2021; Wu & Trottier, 2021). Face-disclosure and body-disclosure, however, can have very different implications for heterosexual users. In a China-based study, Xu and Wu (2019) show that on Momo, an app that does not entail a matching mechanism, using face pictures in a profile is related to hook-ups, as appearance is vital to sexual attraction. In contrast, men and women whose faces and/or bodies cannot be clearly seen in their profiles tend to relate to each other mentally, with their communication often based on shared interests or similar backgrounds. On the other hand, displaying body parts in profiles seems to be more accepted by gay men than by heterosexual users, as revealed in an Italy-based study (Comunello et al., 2021). According to Comunello and her colleagues, heterosexual female users are not interested in seeing male users showing off their muscles. Meanwhile, heterosexual males tend to morally judge women showing body parts in profiles, perceiving them to be indecent. Lesbians reportedly dislike women presenting themselves in a sexy way online as well.

Interactions between users

During their interactions with others, users may signal their interactive goals, discern others' goals, and negotiate a common goal. Certainly, interaction patterns on dating apps are related to users' motives. Licoppe et al. (2016) demonstrate how Grindr users in France adopt communication strategies to arrange immediate casual sex. Notably, Grindr users have evolved a particular 'linguist ideology' which informs them of the association between conversation types and interactive goals. They know that an ordinary 'friendly' conversation that includes sharing personal information and experiences can lead to relational development, thus unsuitable to the immediate sexual purpose. Therefore, they avoid referring to personal issues and biographical details that could lead to greater social and emotional involvement. They routinize the chat and follow a particular sequence, seeming to have a checklist in their mind that they can quickly run through. This checklist-style talk is sex-oriented. It has several features. First, users ask and answer questions in a way that relevant information (e.g. personal pictures, location, immediate sexual goal) is explicitly and briefly revealed. Second, at the beginning of the talk, questions can be raised rapidly one after another, leaving the interrogee little time to reply to each in turn. Third, one may take the initiative to send personal pictures and share the location, aiming to encourage reciprocity.

Resonating with Licoppe and his colleagues, Wu and Ward (2020) have also noted that gay users are sensitive to different types of conversations on dating apps. Focusing on Chinese gay singles in Beijing, they reveal that relationship development on dating apps is often driven by the 'casual conversation,' which does not revolve around clearly articulated pragmatic purposes. Instead, it tends to

unfold around personal experiences and common hobbies. For users who enjoy casual conversations, two other types of conversations are highly instrumental and undesirable. The first type is the sex-oriented conversation, which is aimed at immediate sexual encounters and often starts with impersonal sex-related questions about interactive intentions, body features, and sexual preferences. The other type is the interrogative conversation, in which people ask personal questions (about age, physical traits, profession, hobbies, romance history, etc.) in a nonreciprocal, rigid way. It seems that the purpose of the interrogative conversation is to quickly judge if one is suitable as a dating partner. Wu and Ward (2020) categorize both the sex-oriented conversation and the interrogative conversation into the 'pragmatic conversation.' They argue that pragmatic conversations stifle 'sociability,' or the pleasure of socializing itself, in online dating and are unlikely to contribute to relationship development. Interestingly, this craving for sociability in interactions is also found in Chan's (2020) study on Chinese heterosexual users, framed as 'anti-purposefulness' by Chan. While Chan's interviewees have relatively high socio-economic statuses, Liu et al. (2021) focus on disadvantaged Chinese rural migrant workers. The study shows that female workers often have materialistic expectations for their potential partners, and male workers often receive enquiries about their economic conditions (e.g. 'Do you own a house?') from female users in geographical proximity who are most likely factory workers themselves. This contradicts the condition for sociability craved by the presumably urban, middle-class users.

When we look at interactions in heterosexual cases, gender differences are clear. Comunello et al. (2021) show that it is the men who often take the initiative to start a conversation after getting a match on Tinder rather than women. It is also the men who often try harder to keep the conversation going on. Moreover, some hold a stereotypical view that men tend to 'swipe right' on more profiles than women do; men are eager to enhance their chance of getting a match, while women are more careful. Comunello et al. argue that these findings are consistent with traditional gender scripts in society: Men are assumed to be assertive and interested in sex, while women are expected to be reactive and value emotional needs over sex.

Interactions on dating apps do not always have a happy ending. For instance, one party may cut off contact with the other and ignore their attempts to reach out. This is commonly called 'ghosting,' and it can be a painful experience for users (Timmermans et al., 2020).

Choosing among apps

There are many dating apps on the market for users to choose from. While many studies focus on one app and its users, some aim to examine how users differentiate various apps and choose certain ones for their dating practices. The latter show that in the selection process, users normally consider how the apps can be used and what kind of people they can connect with on the apps.

In a London-based study, MacKee (2016) demonstrates that gay men have appropriated the mainstream app Tinder as a platform suitable for 'serious' dating while maintaining other hyper-sexualized dating apps (e.g. Grindr) as venues for exploring their sexual desires. In the local gay community discourse, Tinder is constructed as a site where 'quality' gay men who are interested in finding romance gather. This partly results from the design features of Tinder. At the time when MacKee conducted his study, Tinder users had to log in with their Facebook accounts. Their personal data, such as users' names and profile pictures, was directly imported into Tinder from Facebook. In general, gay men's profile pictures on Facebook are not as sexualized as those on gay platforms. This shaped the norms of self-presentation on Tinder. Besides, Tinder also invited users to link profiles to their Instagram accounts, which was an additional way to share content. Tinder introduced an identity verification mechanism that discourages users from either conducting or presenting themselves in a hyper-sexualized manner. To highlight his findings, MacKee has provided an example of a gay man who was using Tinder and Grindr at the same time. With a shirt on in his profile picture, this man mainly mentioned his personality in his Tinder profile. However, his profile picture on Grindr showed him as topless; his profile also emphasized his sexual preferences

as a 'hard top.' As MacKee further shows, desexualization on Tinder also occurs in private chats. While questions about sexual preferences are very common on other apps like Grindr, they rarely arise in gay men's conversations on Tinder. Given that Tinder is socially constructed as a less sexualized space in comparison with gay-specific dating apps, gay men are choosing to focus on romantic quests by choosing Tinder.

While people use different apps for different interactive purposes, as shown in MacKee's (2016) study, they may also expect to connect with a certain group of users by choosing a certain app. Focusing on the Chinese diaspora in Australia, Chen and Liu (2021) reveal how different groups of users are clustered around Tinder and OkCupid, two Western apps, and Tantan, a Chinese app. Specifically, they have identified three groups of ethnic Chinese users. The first group consists of Chinese women who prefer to date white men. In those women's eyes, white men are more accepting of the women who are stigmatized in the Chinese community, such as those who are divorced or who have exceeded the 'marriageable' age. Compared with Chinese men, white men are seen as more tolerant of unconventional sexual practices, such as non-monogamous relationships. They are also seen as more accepting of the female appearances that are less attractive by Chinese standards. Accordingly, female users in the first group tend to use Tinder and OkCupid, whose users are perceived to be dominantly white. The second group are Chinese women who prefer to date Chinese men. These women often do not find white men physically attractive. For them, white men may have an intimidating hypersexual image. Meanwhile, they may want to maintain a sense of belonging to Chinese communities. Some find it tiring to speak English all the time and thus prefer to date Chinese men. Accordingly, they choose Tantan, whose users are dominantly Chinese. Lastly, Chen and Liu have categorized Chinese men into a third group. Chinese men in Australia overwhelmingly prefer to use Tantan and date Chinese women. This is related to the situation that Asian men's sexualities have been constantly been stereotyped and marginalized in the West, which these Chinese men are aware of. While facing inequality in the Australian dating market, Chinese men also tend to question the attractiveness of white women regarding either physical appearances or personalities. They may also perceive white women as 'promiscuous' and regard white-dominated Tinder as more 'sexual' than Tantan. For Chen and Liu, ethnic Chinese users' dating experiences are influenced by racialized and gendered perceptions of digital dating.

Synthesizing the perspectives on both the differentiation of interactive purposes on different dating apps and users' preferences for certain groups of people, Wu and Trottier (2021) demonstrate how metropolitan middle-class gay men in China associate dating apps with different user motives and user groups. They examine the comparisons made by their participants between four dating apps: Aloha, Blued, Grindr, and Tinder. Both developed by Chinese companies, Aloha and Blued are often compared with each other. The matching mechanism of Aloha resembles that of Tinder, which means that profile pictures clearly showing faces are crucial in getting a match. Aloha also affords social feeds, inviting users to visually present themselves like on Instagram. Accordingly, showing one's face has become a norm on Aloha. This leads to the exclusion of MSM from lower social classes or from smaller cities, townships, or rural areas, who are more vulnerable than metropolitan middle-class gay men to the possible negative consequences of revealing their sexual identities. In comparison, Blued has gained the largest user base in the heavily guarded Chinese internet industry, partly due to its market positioning. Without a matching mechanism, it has attracted lots of users who would rather send face photos via private chat instead of showing their faces in profile pictures. Accordingly, Blued has a more diverse user group than Aloha does. Searching for their own kind, metropolitan middle-class gay men thus prefer Aloha to Blued. Regarding the association between these two apps and user motives, the distance-sorted display of nearby users contributes to the predominance of immediate hook-ups on Blued; the matching mechanism of Aloha, however, functions as a 'speed bump' and thus nourishes users' expectations for lasting connections. Craving more intimacy even in casual sex, metropolitan middle-class gay men tend to have a negative perception of impersonal immediate hook-ups and people who engage in it. They tend to

evaluate the average ‘quality’ of Blued users as ‘low’ and that of Aloha users as ‘high,’ also because of other factors such as the difference in dominant aesthetics of self-presentations on the two apps.

On the other hand, Wu and Trottier (2021) reveal that a small group of metropolitan middle-class gay men in China are using the Western apps Grindr and Tinder. Compared to Aloha and Blued, Grindr and Tinder are less successful among Chinese gay men because of their unstable internet connection in China and their absence in the app stores of the smartphones that take up the most Chinese market share (e.g. Huawei, Oppo, Vivo, and Xiaomi). Consequently, Chinese gay users in mainland China who have heard about and have access to Grindr and Tinder are those who have transnational experiences (e.g. studying, working, or travelling abroad) or international horizons, who use iPhones or can access Google Play via a VPN. These users tend to have a relatively high socioeconomic status and thus can be perceived to have ‘high quality’ by others. However, the controversy stems from the fact that foreign travellers or expatriates also tend to use Grindr or Tinder. Some may think that Chinese users of Grindr or Tinder have a racial preference for (white) foreigners. Indeed, those who prefer to date foreigners, for either bio-racial preference or socio-economic cause, may tend to gather around Grindr or Tinder; a similar phenomenon can be seen in a Japan-based study as well (Baudinette, 2019). According to Wu and Trottier (2021), differentiation between the two Western apps also exists: Grindr is more associated with immediate hook-ups, and Tinder with a more lasting connection. Overall, Wu and Trottier (2021, p. 11) demonstrate how ‘design features of dating apps, marketing strategies of app companies, and local internet regulations’ can shape users’ choices by ‘unevenly distributing the platform access to users across social classes and territorial divisions and (dis)enabling particular communicative practices in collective sexual life to different extents.’

Social arrangements and their reformation

The proposition about ‘liquid love’

Many societies share a dystopian view that new media technologies are ruining close relationships (Baym, 2015). Even though people do not always explicitly articulate such a causal relation, they may still suggest a correlation between new media and the erosion of durable intimacy. In academia, a representative of this pessimistic view is Bauman’s proposition about ‘liquid love’ (Bauman, 2003). Bauman argues that love seems to be less solid in ‘the liquid modern world that abhors everything that is solid and durable’ (Bauman, 2003, p. 29). People who seek a sense of solidarity and security in romantic love are also troubled by the ‘perpetual uncertainty’ (Bauman, 2003, p. 14): whether they have chosen the right person and done the right thing by being committed to a relationship. The feeling of uncertainty can be intensified by the abundance of romantic alternatives, making people less willing to ‘settle down.’ New media technologies play a role in this process, as they can easily connect people to a large pool of romantic alternatives. Endorsing the metaphor of ‘shopping’ for online dating, Bauman seems to suggest that online daters are less likely to be committed to a relationship, since the internet provides them with endless options.

Consciously or not, many dating app researchers are responding to the proposition about ‘liquid love.’ The presupposition is that users often experience an abundance of potential romantic partners on dating apps, though on some occasions one may experience scarcity instead (Duguay, 2019). In an Australia-based study, Hobbs et al. (2017) explore whether dating apps are facilitating ‘liquid love,’ as the users can access ‘an extensive network of romantic possibilities’ (Hobbs et al., 2017, p. 281). According to their findings, dating app users indeed feel they have more dating opportunities than previous generations, and that dating apps give them greater agency in pursuing and meeting potential romantic partners. However, dating apps are by no means equally beneficial to all users. Some feel that the possibilities offered by dating apps are still quite limited, and only attractive people can fully take advantage of the extensive network. Some think that the connections on dating apps are mostly based on profile photos and thus are superficial. Overall, most users seem not

to be 'spoiled' by the extensive network of romantic possibilities. They still value traditional ideals of monogamy, commitment, and the notion of romantic love, and use dating apps to pursue meaningful partnerships. Indeed, as Bosch (2020) argues in a study based in South Africa, there is little evidence that 'the networks of romantic possibility offered by online dating is eroding traditional ideals of commitment and romantic love' (p. 8). Instead of being spoiled, users may fear being single, lack self-esteem, and experience partner choice overload if many potential partners are available (Thomas et al., 2022).

Other studies are also informed by a choice overload on dating apps and the metaphor of 'relationshipshopping' (Alexopoulos et al., 2020; Sharabi & Timmermans, 2021). For example, Sharabi and Timmermans (2021) examine the association between online dating intensity and commitment in a quantitative study based in an American university. This association, as they have hypothesized, is mediated by the quality of romantic alternatives, investments, and satisfaction with the online dating relationship. Their findings show that those who use dating apps more intensively tend to be more committed to their online dating partners. As the authors explain, people who invest more time and effort in dating app use may have a greater interest in securing a relationship. Meanwhile, compared to the investments and satisfaction, the quality of alternatives is a weaker moderator of the association between online dating intensity and commitment. The researchers thus point out that we should take into consideration dating app users' estimation of the possibility of obtaining an alternative partner. In fact, lots of people have experienced frustration in dating app use. Even with the abundance of potential dates, they may still feel it is hard to find the right person. Finally, Sharabi and Timmermans (2021) have also noticed that users with higher online dating intensity are less willing to stop using dating apps when they start dating someone regularly. As for those who are eager to establish a romantic relationship, they may also tend to uphold the ideals about a romantic partner and be more cautious in ascertaining if they have met the right person.

The above-mentioned studies mainly explore whether dating apps influence users' willingness to *get into* a committed relationship. On the other hand, some studies are concerned with existing relationships, associating dating app use with infidelity. In a study based in the United States, Weiser et al. (2018) point out that non-single users are using Tinder for meeting extradyadic partners, and people overwhelmingly view Tinder as a method for engaging in infidelity. In comparison, Timmermans et al. (2018) reveal a more complicated picture regarding non-single users' motives, although they have not specified which country they collected the data in. As mentioned earlier, they show that non-single Tinder users are not necessarily seeking sexual or long-term encounters; they may simply want to satisfy their curiosity about the current dating market and understand their own values as potential dating partners. Nevertheless, the researchers still compared the personality traits of non-single users with those of single users and partnered non-users, implying that non-single users may be less socially desirable in general.

The association between dating apps and infidelity seems arbitrary. Not only because non-single users are not necessarily using dating apps for extradyadic sexual or romantic involvement, but also because shared understandings of infidelity are often confined by heteronormative monogamous perspectives. Challenging heteronormativity, Møller and Petersen (2017) focus on the appropriation of dating apps in gay men's non-monogamous relationships. In the British cases they present, non-monogamous gay men use dating apps for sexual or more intimate relations while remaining within the negotiated boundaries of their relationships. Møller and Petersen identify two intimacy scripts at work in these relationships. One script is referred to as 'monogamous intimacy,' wherein the couple reserve emotional exclusivity while extradyadic sex is allowed. The other is 'non-monogamous intimacy': As the boundaries between friendships, romantic relationships, and sexual encounters are blurred, neither emotional exclusivity nor sexual exclusivity is required in the relationship. Dating apps fit into both scripts without implicating infidelity in the monogamous sense.

Møller and Petersen's (2017) perspective, according to Wu (2021), has its limitations. Focused on non-monogamous gay couples and the sexual affordances of dating apps, this perspective dismisses the apps' less sexual role as a connection to the gay community, which can be important to non-

single gay men in either monogamous or non-monogamous relationships. Responding to this research gap, Wu (2021) explores how non-single gay men and their partners in China 'domesticate' dating apps, or, in other words, how dating apps can be appropriated and accepted in gay romantic relationships. Wu highlights dating apps' dual role as a pool of sexual/romantic alternatives and a channel to the gay community. While the former may disturb a romantic relationship, the latter leaves room for negotiation between a couple for acceptable but restricted usage. Embedded in relational dynamics, domestication of dating apps is achieved through negotiations of relationship members and much defined by available relationship scripts. When negotiating over dating app use, gay couples are also negotiating the relational boundaries and norms. Accordingly, domestication of dating apps can result in either the reinforcement of monogamy or the embrace of non-monogamy. As for the symbolic and the cognitive dimensions, Wu (2021, p. 528) writes:

Despite dating apps' dual role in actual practices, domestication on the symbolic dimension entails monogamous couples' intentional construction of an unremarkable image of dating apps. Monogamous Chinese gay men achieve that by dismissing dating apps as unremarkable as other social media platforms and putting their faith in user agency. This often involves a cognitive process where they learn to analyze the relationship experience of themselves or others, with their perspectives often being sociological or psychological, and debunk the arbitrary association between dating apps and infidelity. However, they may also go through another cognitive process where they gradually embrace the notion of non-monogamy, accepting the sexual or even the romantic affordances of dating apps. In that case, they are even less likely to attach negative symbolic meanings to dating apps and regard dating apps as a threat.

Overall, Wu (2021) argues that Chinese gay men tend to avoid a deterministic view regarding dating apps' influence on romantic relationships. This message can be interesting for researchers who often engage in discussions on technological determinism, social constructivism, and the mutual shaping thesis. Particularly, it reminds us that it is worth researching how a certain view on technology-society relationship is adopted by technology users in their daily lives. For dating app researchers who propose or respond to the proposition about 'liquid love,' it is also a reminder that we should pay attention to how users themselves negotiate the assumed abundance of sexual or romantic alternatives presented by dating apps.

Casual sex and durable relationships

In many studies on gay men's dating app use, researchers have highlighted the sexual affordances of dating apps. For instance, Gudelunas (2012) maintains that dating apps have enhanced gay men's ability to seek casual sexual encounters; Tziallas (2015) attributes the success of gay dating apps partly to their functioning as amateur pornography platforms; Licoppe et al. (2016) demonstrate how Grindr users deliberately circumvent emotional involvement through strategic interaction. Meanwhile, dating apps are constantly referred to as 'hook-up apps' (e.g. Albury & Byron, 2016; Davis et al., 2016; MacKee, 2016; Race, 2015a), which further obscures the role played by dating apps in the development of 'meaningful' social relationships.

Against this backdrop, Yeo and Fung (2018) focus on gay users' experiences of seeking durable relationships in a study based in Hong Kong. They demonstrate that the technological features of dating apps promote a specific tempo and sequence of browsing and interaction, which deviate from the normative temporal patterns in the formation of friendships and romantic relationships. Regarding the tempo, dating apps enable more frequent and efficient connections between users, generating feelings of accelerated relationship formation. While the formation of durable relationships takes a gradual pace, the fast tempo on dating apps tends to result in ephemerality and loss of intimacy. On the other hand, design features of dating apps structure the sequence of browsing in a way that emphasizes users' physical appearances, as the photos are what users first see in others' profiles. The sequence of messaging is not helpful either, as it also registers a 'value hierarchy that prioritizes physical attractiveness and devalues the exploration of mind and character' (Yeo & Fung, 2018, p. 12). Overall, the temporality of browsing and interactions on dating apps seems

to privilege casual sex instead of durable relationships. Therefore, users who hope to find friends or romantic partners are often frustrated.

Like Yeo and Fung, Wu and Ward (2020) are also interested in gay users' experiences of relationship development. In their Beijing-based study, they explore how these experiences are shaped by gay users' negotiation of the relation between casual sex and 'serious' relationships, and by the affordances of dating apps. Their findings suggest that casual sex is not always at odds with relationship development. Metropolitan single gay users have 'relationalised' casual sex by perceiving it as a form of social connection and endowing it with the potential to foster a relationship. In other words, casual sex also satisfies users' emotional demands to some extent and can be the start of a durable relationship. Accordingly, these single gay users prefer 'relationalised' casual sex to cold, impersonal casual sex. Open to both casual sex and durable relationships, they seek to connect with 'interesting people' who can have a good conversation with them. From this connection, they derive 'sociability,' or the pure pleasure in socializing with others. Dating apps' affordance of stranger connectivity provides them with abundant potential connections. While relationship development is accompanied by people's needs for communicating more synchronously and getting to know more about each other, dating apps have a relatively low degree of communicative synchronicity and limited user identifiability. Therefore, when the relationship between two users develops, they tend to switch to the mainstream social media platform WeChat. Platform switching as a milestone in relationship development is also discussed in Broeker's (2021) research.

Yeo, Fung, Wu, and Ward situate gay men's dating app use in the pursuit of social relationships that are recognizable to the general public. In contrast, Race (2015a, 2015b) takes 'cruising' – searching in public places for sexual partners, which is a long-standing practice among men who have sex with men – as a reference point for understanding the transformation of gay sexual practices in the rise of dating apps. Traditionally, cruising happens in places like public restrooms and parks. At a cruising venue, men often engage in sex encounters silently, although exceptions exist (Tewksbury, 2008). In comparison, a preceding chat process is indispensable on dating apps. According to Race (2015b), chat mechanisms afforded by dating apps enable multiple forms of controlled and anonymized self-disclosure of, for instance, sexual interests and HIV status before sexual encounters, thus forging new modes of partner sorting and risk prevention. Meanwhile, users are enabled to co-construct their sexual fantasies and make arrangements for their upcoming sexual encounters (Race, 2015a, 2015b). Apart from the chat mechanisms, other affordances of dating apps also constitute a transformative force in gay men's sexual practices. Most of all, the capacity to search users, add 'buddies,' and keep track of 'favourite' allow sexual encounters with certain users to reoccur. Given dating apps' functionalities of storage and retrieval, Race (2015b, p. 505) claims that 'the capacity to maintain a loose web of concurrent fuck-buddies is perhaps more available, more accessible and more widely accessed than ever.' Believing that gay men gain affective bonds and affinities in hook-ups through dating apps, Race (2015a, p. 271) says, 'These devices and practices are participating in the construction of a specific sphere of sociability and amiable acquaintance among men in urban centres that prioritizes sex as a principle [*sic*] mechanism for connection and sociability.'

Impact on gay communities

There is a long-running debate about the fate of gay communities that used to be anchored to physical venues (e.g. gay bars, gay neighbourhoods, etc.). The concept of 'gay community' has been especially of interest to HIV prevention researchers, because gay communities have played an important role in HIV prevention work, such as disseminating the knowledge of safe sex (Holt, 2011). Particularly, in the United States academics also value the gay community and offline queer spaces for their anti-heteronormativity role (Renninger, 2019). The rise of the internet and digital devices has triggered the debate on whether gay communities are declining, as the attachment to the (offline) gay community may become less necessary for gay men to connect with each other

(Holt, 2011; Rosser et al., 2008; Rowe & Dowsett, 2008; Zablotska et al., 2012). Unsurprisingly, dating apps are also involved in this debate.

Davis et al. (2016) argue against the nostalgic, monolithic, and metropolitan-centric view on the fate of gay communities, reminding us that it has never been easy for culturally and geographically marginalized gay men to get access to publicly visible gay communities. Their study shows the values of dating apps and websites for gay men in Lanarkshire, a Scottish county. Poor transportation links and a lack of local commercial gay scenes make some gay men there relatively isolated. As violence against gay men haunts Lanarkshire towns and villages, gay men tend to have a safety concern when meeting a stranger in public spaces such as a cruising park. In contrast, dating apps and websites make gay men feel safer, as the exchange of some identity information creates assumed reciprocity and generates trust. Meanwhile, an HIV-positive gay man may also feel easier and safer to disclose his status on digital media. Therefore, the researchers argue:

Websites and apps may figure as pernicious in the discourse of those who are comfortable in their social worlds and see no use for them, whereas other men see digital media differently. A key point of this argument, therefore, is that digital media look different depending on one's social location – geographic, health status, income, class, appearances and otherwise – as a gay man. (Davis et al., 2016, p. 847)

In another study, Renninger (2019) challenges the 'Grindr is killing the gay bar' argument by calling for attention to many other factors that facilitate the changes of 'gayborhoods' in American cities. As Renninger argues, people are ignoring the political economy of urban development when they blame Grindr for the closing of gay bars. Gayborhoods and gay spaces are subject to, for instance, the process of gentrification in urban development. Renninger also demonstrates that Grindr is not a replacement for gay bars, given the ways people use the app. As it is not rare for gay men to use Grindr while being in a gay bar, the online and offline gay spaces can overlap and get entangled, forging a form of co-existence. Regarding the value of Grindr to the gay community, Renninger (2019, p. 1749) maintains that 'Grindr creates connections and contact across neighbourhoods, a phenomenon that is valuable to many as gayborhoods deteriorate and decentralize.' Notably, he is not the only scholar who recognizes dating apps as an important bridge to a larger community for LGBTQs (e.g. Ong, 2017; Pym et al., 2021; Wu, 2021).

Based on fieldwork in Japan, Baudinette (2019) argues that gay dating apps have reinforced, not eroded, physical queer space. His study focuses on Tokyo's gay town of Shinjuku Ni-chōme, which remains an important space for gay men's socialising. He argues that Ni-chōme exists as 'both a real, physical space and a virtual, imagined space that is accessible via gay dating applications and social media services' (p. 93). When a gay man is physically in Ni-chōme, using dating apps and other social media to communicate with other gay men can foster a sense of shared community. In addition, by displaying a high concentration of gay men in this area, dating apps can enhance gay men's knowledge of Ni-Chōme as a gay town, and can lead curious gay users to Ni-Chōme, including those who are unaware of Ni-Chōme's status as a gay town. Therefore, Baudinette (2019, p. 103) argues that the virtual connectivity afforded by dating apps has reinforced Ni-chōme's position as 'a space awash with queer affective potential.'

Relational homophily

Although media and communication researchers often expect to see how new digital media facilitate the reformation of social relations, established patterns in human behaviour sometimes remain unchallenged. One pattern that still holds up in the rise of dating apps is relational homophily, or people's tendency to be attracted to those who are 'similar' to themselves (Parisi & Comunello, 2020). Similarities are found across measures including education background (Ranzini et al., 2022), personality (Neyt et al., 2020), and political stance (Chan, 2021a).

Algorithms may have contributed to relational homophily in the case of dating apps by catering to our preferences. Many dating apps present users with preselected potential partners or

recommend ‘best’ matches, with algorithms playing a role in this process (Comunello et al., 2021). One type of algorithm that is possibly employed by dating apps is the collaborative filtering algorithm, which has been used to build recommendations for users by online platforms such as Amazon and Google (Nader, 2021). Nader (2021) argues that collaborative filtering algorithms, if employed by dating apps indeed, can homogenize and deepen users’ exclusionary preferences. He writes:

The idea behind collaborative filtering is that if groups of users show similar patterns of preferences, the preferences of one user can be predicted from the past behavior of similar users. In other words, by collecting data on the preferences of users collectively, the algorithm predicts the preferences of an individual user, builds recommendations that match those preferences, and filters the content the user can access on the platform. (Nader, 2021, pp. 238–239)

Accordingly, it is the collective patterns of behaviour, not individual preferences, that are prioritized to predict individual users’ preferences. In the case of dating apps, users that are grouped together by collaborative filtering algorithms will receive the same recommendations and thus are likely to match with the same people. This may amplify the existing biases against disadvantaged groups, such as racial biases (Nader, 2021).

Nader’s study is enlightening for us. If we assume that individual users’ preferences for similar others can be reflected in collective data and captured by collaborative filtering algorithms, it is reasonable to believe that these algorithms can also enhance relational homophily.

Sexual racism

Dating app researchers seem to agree that sexual racism – the racial discrimination people face in sexual or romantic contexts – is reproduced through dating apps. In the mobile dating context, sexual racism is manifest either in a blatant, aggressive form or in a subtle, covert manner; it is encountered by both heterosexuals and LGBTQs in different societies (Ang et al., 2021; Carlson, 2020; Hanson, 2021; Lee, 2021; Li & Chen, 2021; Mason, 2016; Peck et al., 2021; Stacey & Forbes, 2021).

Dating apps are by no means race-neutral. Either the ways they are marketed or their design features may carry racial connotations (Bivens & Hoque, 2018; Duguay, 2017; Shield, 2019). For instance, Duguay (2017) points out that featured actors in Tinder’s marketing are prominently white. Bivens and Hoque (2018) reveal that Bumble was targeted at white heterosexual women in the beginning. As Shield (2019, p. 203) maintains, Grindr is exporting an American racial logic internationally and can be seen as ‘part and parcel of U.S. American cultural imperialism.’ Since Grindr is an app developed in the United States and has an ‘ethnicity’ drop-down menu that invites users to provide racial or ethnical information in their profiles, the options under this menu (e.g. Asian, Black, Latino, White, etc.) are largely embedded in dominant American notions of race and ethnicity. Meanwhile, Shield (2019) has noticed that queer male migrants to the West may ‘learn’ to identify as one of the ethnicities pre-set by Grindr, often a label that does not make sense in his own culture.

The American ethnical categorization and racial hierarchy, however, have not necessarily assimilated their counterparts in other societies through globalized dating apps such as Grindr. In Singapore, for instance, Grindr’s ethnicity drop-down menu is less useful, as most residents there would fall into the broad category of ‘Asian’ or ‘South Asian’ (Ang et al., 2021). Officially, Singaporeans are ethnically categorized into Chinese, Malays, Indian, and Others, with Chinese being the majority. In this context, Ang et al. (2021) find that some Grindr users in Singapore would add local ethnical labels (e.g. ‘local Chinese’) in their profiles. They also find that local Chinese users are generally considered the most desirable in the Singaporean gay dating scene. Responding to this racial hierarchy, racial minorities may take strategies such as ‘trying to present a Chinese or Chinese-mixed racial identity, emphasizing an identity

that is globalized rather than ethnic, and reframing their situation to disavow their victimhood' (Ang et al., 2021, p. 129).

Gender relations

Some studies examine how dating apps reproduce problematic gender relations. Bivens and Hoque (2018) show that the dating app Bumble, despite its self-declared feminist social justice objective being gender equality, still operates on the stereotypical assumptions about gender, sex, sexuality. Like Tinder, Bumble works with a matching mechanism: If two users indicate an interest in one another by swiping right, they get a match and then can chat through private messaging. Aware that many female users of dating apps have received unsolicited explicit pictures or aggressive messages from men, the creators of Bumble added a design feature to Bumble to solve this problem and distinguish it from other mainstream dating apps: the 'lady ask first' feature. It means that only female users can start a conversation, within 24 h after the match occurs. The match will expire if a conversation is not initiated. This design feature is supposed to allow women to stay in control. However, it also reinforces the stereotypical perceptions of genders. As Bivens and Hoque (2018, p. 449) argues:

The logic of Bumble relies on policing the male body as if it always exists in an intrinsically brusque and brash state of being, only capable of performing a savage performance of masculinity. Through this lens, males are by nature uncivil and incapable of comportment. This is a very narrow construction of masculinity. Meanwhile, female bodies and feminine characteristics are situated as more genteel.

According to Bivens and Hoque (2018), Bumble's design is also problematic in other aspects. For instance, when setting up profiles, Bumble users are only offered two gender options, male and female, leaving out other categories such as transgender and cisgender. The 'ladies talk first' feature fails to work for lesbian and gay users, who were not taken into consideration when Bumble was first developed; the gender assumptions which this feature is based on are also racialized. In a nutshell, Bivens and Hoque (2018) argue that the intended users of Bumble from the beginning were limited to white cisgender heterosexual women.

In another study, MacLeod and McArthur (2019) present an interface analysis of Tinder and Bumble, revealing that the two apps construct gender as a rigid category to serve the matching process instead of one's gender performance. Providing only two gender options, male and female, the profile settings of these apps do not account for a full range of user identities. Besides, the gender setting is not publicly displayed, and it is more about what types of profiles the app should present to the user. For instance, a person who has selected 'female' for gender and 'only women' for sexual preference will be presented with other 'female' users who have chosen 'only women' or 'men and women.'

Apart from how the app infrastructure reproduces stereotypical assumptions about sex, gender, and sexuality, researchers also show interest in problematic gender performance in users' interactions. Notably, heterosexual female users may encounter men's intrusive behaviour – such as harassment and even sexual violence in app-facilitated meetings – and need to take a range of online and offline strategies to feel safe (Chan, 2018; Gillett, 2021; Pruchniewska, 2020; Thompson, 2018). Hess and Flores (2018) have identified three types of heterosexist performances of 'toxic masculinity' in male users' pickup lines in an English-speaking context on Tinder. The first type is groan-inducing failed pickup lines, through which many users express hypermasculinity and make poor attempts to hit on women. The second is hypersexual declarations, featured with the overt sexualization of the female body. The last type is objectification through consumption. It refers to the use of food or consumption metaphors, which equates the consumption of food with the consumption of women. On the other hand, Hess and Flores have also paid attention to how the women who find these pickup lines annoying. While many would strategically remain silent, others directly respond, either through an improvisational style or by calling out the failed pickup line for what it is.

Conclusion

Dating app studies offers flourishing insight into dynamics between apps and social settings, including macro-level contexts (e.g. the national and the cultural) and micro-level situations (e.g. the daily and the individual). We see that the reconfiguration of dating apps is perhaps subject to more internet regulations in China than in Western countries, and the Chinese dating apps have taken up more functionalities than their Western counterparts. We see that dating apps operate on and in turn reproduce a sense of community for LGBTQ users in particular. We also see that the sexual and romantic affordances of dating apps assume distinct meanings in monogamous and non-monogamous relationships, given the different relational boundaries. Overall, it is in those well-contextualized studies that we find the most enlightening knowledge about dating apps. We believe that research on dating apps can yield more insight by diving into the socio-cultural and daily relational contexts that are underexplored, or re-examining those we take for granted.

For instance, regarding the reconfiguration of dating apps, studies on Chinese dating apps address the role of internet regulations, reminding us that political entities such as nation-states remain a relevant, significant point of reference. In contrast, we do not see the kinds of internet regulations Western apps encounter, at least in the studies we found. This does not imply that there are no internet regulations in Western countries. Instead, it is probably because internet regulations are taken for granted when researchers explore the reconfiguration of Western dating apps. For better contextualization, dating app researchers should not dismiss governmental interventions in countries where internet companies are often assumed to have more freedom. The significance of this perspective already can be seen in the case of Grindr: In 2019, the U.S. government forced Beijing Kunlun Tech – a Chinese gaming and technology company that owned Grindr at the time – to sell Grindr in the name of ‘national security’ (Platt et al., 2019).

Given that dating apps are first developed in a specific socio-cultural context, it would be interesting to examine whether and how dating apps are reconfigured as they gradually become globalized. Studies have already shown that design features of dating apps are hardly suitable for all socio-cultural contexts, such as the ‘ethnicity’ drop-down menu of Grindr (Ang et al., 2021; Shield, 2019). Accordingly, we may ask whether or to what extent dating app developers consider users from other societies. In other words, are the developers of those globalized dating apps able to sustain a trans-cultural vision? How do they react to the feedback from international users, as well as the internet regulations in other countries, when their products keep expanding on the global market? How might they reconfigure their apps to reflect a cosmopolitan spirit? On the other hand, researchers may also explore the cases in which app developers consciously focus on a local market and analyse their strategies for localization, if such cases exist.

Regarding user practices and their remediation, we expect to see more research on the currently understudied user groups. For instance, most studies attend to heterosexual and gay male users, with bisexuals, lesbians, transgenders, and other queer users receiving less attention. Users with religious backgrounds are not fully examined either. Besides, users discussed in existing research are relatively young, which leads to an age-related gap in the literature. We also notice that many researchers in the United States tend to recruit research participants from university students. Thus, we urge those researchers to get out of their comfort zone and pay more attention to users of other age groups and/or social classes. In addition, existing studies are mostly focused on users in Western countries and China. We hope to see more studies based elsewhere in the future.

Social arrangements and their possible reformation also warrant more scrutiny and need to be discussed in a contextualized manner, given that societies differ in their internet policies, norms about intimacy, gender relations, ethnic relations, etc. Consider Grindr and its implications for ethnic relations as an example. We already know that Grindr is exporting American notions of race and ethnicity, which can be very different from those in other countries (Ang et al., 2021; Shield, 2019). In China alone, for instance, there are 56 officially recognized

ethnicities (Mullaney, 2011), far more than the ethnic categories Grindr has listed. We should ask how Grindr users in non-U.S. contexts accept, appropriate, or reject American racial and ethnic logics, or how the latter may shape their understanding of the ethnic relations in their own societies.

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References

- Albury, K., & Byron, P. (2016). Safe on my phone? Same-sex attracted young people's negotiations of intimacy, visibility, and risk on digital hook-up apps. *Social Media + Society*, 2(4), 1–10. <https://doi.org/10.1177/2056305116672887>
- Alexopoulos, C., Timmermans, E., & McNallie, J. (2020). Swiping more, committing less: Unraveling the links among dating app use, dating app success, and intention to commit infidelity. *Computers in Human Behavior*, 102, 172–180. <https://doi.org/10.1016/j.chb.2019.08.009>
- Ang, M. W., Tan, J. C. K., & Lou, C. (2021). Navigating sexual racism in the sexual field: Compensation for and disavowal of marginality by racial minority Grindr users in Singapore. *Journal of Computer-Mediated Communication*, 26(3), 129–147. <https://doi.org/10.1093/jcmc/zmab003>
- Baudinette, T. (2019). Gay dating applications and the production/reinforcement of queer space in Tokyo. *Continuum*, 33(1), 93–104. <https://doi.org/10.1080/10304312.2018.1539467>
- Bauman, Z. (2003). *Liquid love: On the frailty of human bonds*. Polity.
- Baym, N. K. (2015). *Personal connections in the digital age* (2nd ed.). Polity.
- Berker, T., Punie, Y., & Ward, K. (Eds.). (2006). *Domestication of media and technology*. Open University Press.
- Bivens, R., & Hoque, A. S. (2018). Programming sex, gender, and sexuality: Infrastructural failures in 'feminist' dating app Bumble. *Canadian Journal of Communication*, 43(3), 441–459. <https://doi.org/10.22230/cjc.2019v44n3a3375>
- Bonner-Thompson, C. (2017). 'The meat market': Production and regulation of masculinities on the Grindr grid in Newcastle-upon-Tyne, UK. *Gender, Place & Culture*, 24(11), 1611–1625. <https://doi.org/10.1080/0966369X.2017.1356270>
- Bonner-Thompson, C. (2021). Anticipating touch: Haptic geographies of Grindr encounters in Newcastle-upon-Tyne, UK. *Transactions of the Institute of British Geographers*, 46(2), 449–463. <https://doi.org/10.1111/tran.12417>
- Bosch, T. E. (2020). *Social media and everyday life in South Africa*. Routledge. <https://doi.org/10.4324/9780429316524>
- Broeker, F. (2021). 'We went from the anonymity of the internet into my private WhatsApp': Rituals of transition among dating app users in Berlin. *New Media & Society*, 14614448211029200. <https://doi.org/10.1177/14614448211029200>

- Carlson, B. (2020). Love and hate at the cultural interface: Indigenous Australians and dating apps. *Journal of Sociology*, 56(2), 133–150. <https://doi.org/10.1177/1440783319833181>
- Chan, L. S. (2017a). The role of gay identity confusion and outness in sex-seeking on mobile dating apps among men who have sex with men: A conditional process analysis. *Journal of Homosexuality*, 64(5), 622–637. <https://doi.org/10.1080/00918369.2016.1196990>
- Chan, L. S. (2017b). Who uses dating apps? Exploring the relationships among trust, sensation-seeking, smartphone use, and the intent to use dating apps based on the integrative model. *Computers in Human Behavior*, 72, 246–258. <https://doi.org/10.1016/j.chb.2017.02.053>
- Chan, L. S. (2018). Liberating or disciplining? A technofeminist analysis of the use of dating apps among women in urban China. *Communication, Culture and Critique*, 11(2), 298–314. <https://doi.org/10.1093/ccc/tcy004>
- Chan, L. S. (2020). Multiple uses and anti-purposefulness on Momo, a Chinese dating/social app. *Information, Communication & Society*, 23(0), 1515–1530. <https://doi.org/10.1080/1369118X.2019.1586977>
- Chan, L. S. (2021a). Looking for politically like-minded partners: Self-presentation and partner-vetting strategies on dating apps. *Personal Relationships*, 28(3), 703–720. <https://doi.org/10.1111/pere.12375>
- Chan, L. S. (2021b). *The politics of dating apps: Gender, sexuality, and emergent publics in urban China*. MIT Press.
- Chen, X., & Liu, T. (2021). On ‘never right-swipe whites’ and ‘only date whites’: Gendered and racialised digital dating experiences of the Australian Chinese diaspora. *Information, Communication & Society*, 24(0), 1247–1264. <https://doi.org/10.1080/1369118X.2019.1697341>
- Chin, K., Edelstein, R. S., & Vernon, P. A. (2019). Attached to dating apps: Attachment orientations and preferences for dating apps. *Mobile Media & Communication*, 7(1), 41–59. <https://doi.org/10.1177/2050157918770696>
- Comunello, F., Parisi, L., & Ieracitano, F. (2021). Negotiating gender scripts in mobile dating apps: Between affordances, usage norms and practices. *Information, Communication & Society*, 24(8), 1140–1156. <https://doi.org/10.1080/1369118X.2020.1787485>
- Davis, M., Flowers, P., Lorimer, K., Oakland, J., & Frankis, J. (2016). Location, safety and (non) strangers in gay men’s narratives on ‘hook-up’ apps. *Sexualities*, 19(7), 836–852. <https://doi.org/10.1177/1363460716629334>
- Duguay, S. (2017). Dressing up Tinderella: Interrogating authenticity claims on the mobile dating app Tinder. *Information, Communication & Society*, 20(3), 351–367. <https://doi.org/10.1080/1369118X.2016.1168471>
- Duguay, S. (2019). ‘There’s no one new around you’: Queer women’s experiences of scarcity in geospatial partner-seeking on Tinder. In C. J. Nash & A. Gorman-Murray (Eds.), *The geographies of digital sexuality* (pp. 93–114). Springer. https://doi.org/10.1007/978-981-13-6876-9_6
- Duguay, S. (2020). You can’t use this app for that: Exploring off-label use through an investigation of Tinder. *The Information Society*, 36(1), 30–42. <https://doi.org/10.1080/01972243.2019.1685036>
- Ellison, N., Heino, R., & Gibbs, J. (2006). Managing impressions online: Self-presentation processes in the online dating environment. *Journal of Computer-Mediated Communication*, 11(2), 415–441. <https://doi.org/10.1111/j.1083-6101.2006.00020.x>
- Fitzpatrick, C., & Birnholtz, J. (2018). ‘I shut the door’: Interactions, tensions, and negotiations from a location-based social app. *New Media & Society*, 20(7), 2469–2488. <https://doi.org/10.1177/1461444817725064>
- Gillett, R. (2021). ‘This is not a nice safe space’: Investigating women’s safety work on Tinder. *Feminist Media Studies*, 1–17. <https://doi.org/10.1080/14680777.2021.1948884>
- Grindr. (2020, June 16). Grindr. <https://www.grindr.com/>.
- Gudelunas, D. (2012). There’s an app for that: The uses and gratifications of online social networks for gay men. *Sexuality & Culture*, 16(4), 347–365. <https://doi.org/10.1007/s12119-012-9127-4>
- Hanson, K. R. (2021). Collective exclusion: How white heterosexual dating app norms reproduce status quo hookup culture. *Sociological Inquiry*, 1–25. <https://doi.org/10.1111/soin.12426>
- Hess, A., & Flores, C. (2018). Simply more than swiping left: A critical analysis of toxic masculine performances on Tinder nightmares. *New Media & Society*, 20(3), 1085–1102. <https://doi.org/10.1177/1461444816681540>
- Hobbs, M., Owen, S., & Gerber, L. (2017). Liquid love? Dating apps, sex, relationships and the digital transformation of intimacy. *Journal of Sociology*, 53(2), 271–284. <https://doi.org/10.1177/1440783316662718>
- Holt, M. (2011). Gay men and ambivalence about ‘gay community’: From gay community attachment to personal communities. *Culture, Health & Sexuality*, 13(8), 857–871. <https://doi.org/10.1080/13691058.2011.581390>
- Jayawardena, M., Pepping, C. A., Lyons, A., & Hill, A. O. (2021). Geosocial networking application use in men who have sex with men: The role of adult attachment. *Sexuality Research and Social Policy*, 19(1), 85–90. <https://doi.org/10.1007/s13178-020-00526-x>
- Kubin, E., & von Sikorski, C. (2021). The role of (social) media in political polarization: A systematic review. *Annals of the International Communication Association*, 45(3), 188–206. <https://doi.org/10.1080/23808985.2021.1976070>
- Lee, J. (2021). Talking through race: Two raced women’s Tinder stories. *Communication, Culture and Critique*, 14(3), 454–470. <https://doi.org/10.1093/ccc/tcab030>
- Lemke, R., & Weber, M. (2017). That man behind the curtain: Investigating the sexual online dating behavior of men who have sex with men but hide their same-sex sexual attraction in offline surroundings. *Journal of Homosexuality*, 64(11), 1561–1582. <https://doi.org/10.1080/00918369.2016.1249735>

- Li, H., & Chen, X. (2021). From “oh, you’re Chinese ...” to “no bats, thx!”: Racialized experiences of Australian-based Chinese queer women in the mobile dating context. *Social Media + Society*, 7(3), 1–11. <https://doi.org/10.1177/20563051211035352>
- Licoppe, C., Rivière, C. A., & Morel, J. (2016). Grindr casual hook-ups as interactional achievements. *New Media & Society*, 18(11), 2540–2558. <https://doi.org/10.1177/1461444815589702>
- Lievrouw, L. A. (2014). Materiality and media in communication and technology studies: An unfinished project. In T. Gillespie, P. J. Boczkowski, & K. A. Foot (Eds.), *Media technologies: Essays on communication, materiality, and society* (pp. 21–51). MIT Press.
- Light, B., Burgess, J., & Duguay, S. (2018). The walkthrough method: An approach to the study of apps. *New Media & Society*, 20(3), 881–900. <https://doi.org/10.1177/1461444816675438>
- Liu, T. (2016). Neoliberal ethos, state censorship and sexual culture: A Chinese dating/hook-up app. *Continuum*, 30(5), 557–566. <https://doi.org/10.1080/10304312.2016.1210794>
- Liu, T., Wang, Y., & Lin, Z. (2021). The cruel optimism of digital dating: Heart-breaking mobile romance among rural migrant workers in South China. *Information, Communication & Society*, 1–18. <https://doi.org/10.1080/1369118X.2021.1874039>
- Mackay, H., & Gillespie, G. (1992). Extending the social shaping of technology approach: Ideology and appropriation. *Social Studies of Science*, 22(4), 685–716. <https://doi.org/10.1177/030631292022004006>
- MacKee, F. (2016). Social media in gay London: Tinder as an alternative to hook-up apps. *Social Media + Society*, 2(3), 1–10. <https://doi.org/10.1177/2056305116662186>
- MacLeod, C., & McArthur, V. (2019). The construction of gender in dating apps: An interface analysis of Tinder and Bumble. *Feminist Media Studies*, 19(6), 822–840. <https://doi.org/10.1080/14680777.2018.1494618>
- Mason, C. L. (2016). Tinder and humanitarian hook-ups: The erotics of social media racism. *Feminist Media Studies*, 16(5), 822–837. <https://doi.org/10.1080/14680777.2015.1137339>
- McGuire, M. L. (2018). The problem of technological integration and geosocial cruising in Seoul. *New Media & Society*, 20(1), 369–383. <https://doi.org/10.1177/1461444816675436>
- Miao, W., & Chan, L. S. (2020). Social constructivist account of the world’s largest gay social app: Case study of Blued in China. *The Information Society*, 36(4), 214–225. <https://doi.org/10.1080/01972243.2020.1762271>
- Miao, W., & Chan, L. S. (2021). Domesticating gay apps: An intersectional analysis of the use of Blued among Chinese gay men. *Journal of Computer-Mediated Communication*, 26(1), 38–53. <https://doi.org/10.1093/jcmc/zmaa015>
- Miller, B. (2015a). “They’re the modern-day gay bar”: Exploring the uses and gratifications of social networks for men who have sex with men. *Computers in Human Behavior*, 51, 476–482. <https://doi.org/10.1016/j.chb.2015.05.023>
- Miller, B. (2015b). “Dude, where’s your face?” Self-presentation, self-description, and partner preferences on a social networking application for men who have sex with men: A content analysis. *Sexuality & Culture*, 19(4), 637–658. <https://doi.org/10.1007/s12119-015-9283-4>
- Miller, B. (2018). Textually presenting masculinity and the body on mobile dating apps for men who have sex with men. *The Journal of Men’s Studies*, 26(3), 305–326. <https://doi.org/10.1177/1060826518773417>
- Miller, B. (2019). A picture is worth 1000 messages: Investigating face and body photos on mobile dating apps for men who have sex with men. *Journal of Homosexuality*, 67(13), 1798–1822. <https://doi.org/10.1080/00918369.2019.1610630>
- Miller, B., & Behm-Morawitz, E. (2016). “Masculine guys only”: The effects of femmephobic mobile dating application profiles on partner selection for men who have sex with men. *Computers in Human Behavior*, 62, 176–185. <https://doi.org/10.1016/j.chb.2016.03.088>
- Møller, K., & Petersen, M. N. (2017). Bleeding boundaries: Domesticating gay hook-up apps. In R. Andreassen, M. N. Petersen, K. Harrison, & T. Raun (Eds.), *Mediated intimacies: Connectivities, relationalities and proximities* (pp. 208–223). Routledge.
- Mullaney, T. (2011). *Coming to terms with the nation: Ethnic classification in modern China*. University of California Press.
- Murray, S., & Ankerson, M. S. (2016). Lez takes time: Designing lesbian contact in geosocial networking apps. *Critical Studies in Media Communication*, 33(1), 53–69. <https://doi.org/10.1080/15295036.2015.1133921>
- Nader, K. (2021). Dating through the filters. *Social Philosophy and Policy*, 37(2), 237–248. <https://doi.org/10.1017/S0265052521000133>
- Neyt, B., Baert, S., & Vandenbulcke, S. (2020). Never mind I’ll find someone like me – Assortative mating preferences on Tinder. *Personality and Individual Differences*, 155, 1–7. <https://doi.org/10.1016/j.paid.2019.109739>
- Ong, J. C. (2017). Queer cosmopolitanism in the disaster zone: ‘My Grindr became the United Nations’. *International Communication Gazette*, 79(6–7), 656–673. <https://doi.org/10.1177/1748048517727177>
- Orosz, G., Benyó, M., Berkes, B., Nikolett, E., Gál, É, Tóth-Király, I., & Bőthe, B. (2018). The personality, motivational, and need-based background of problematic Tinder use. *Journal of Behavioral Addictions*, 7(2), 301–316. <https://doi.org/10.1556/2006.7.2018.21>
- Parisi, L., & Comunello, F. (2020). Dating in the time of “relational filter bubbles”: Exploring imaginaries, perceptions and tactics of Italian dating app users. *The Communication Review*, 23(1), 66–89. <https://doi.org/10.1080/10714421.2019.1704111>

- Peck, A. J., Berkowitz, D., & Tinkler, J. (2021). Left, right, Black, and White: How White college students talk about their inter- and intra- racial swiping preferences on Tinder. *Sociological Spectrum*, 41(4), 304–321. <https://doi.org/10.1080/02732173.2021.1916663>
- Platt, E., Fontanella-Khan, J., & Sevastopulo, D. (2019, March 28). US government is forcing Chinese owners to sell Grindr. *Financial Times*. <https://www.ft.com/content/30408b0e-50e2-11e9-b401-8d9ef1626294>
- Pruchniewska, U. (2020). "I like that it's my choice a couple different times": Gender, affordances, and user experience on Bumble dating. *International Journal of Communication*, 14, 2422–2439.
- Pym, T., Byron, P., & Albury, K. (2021). 'I still want to know they're not terrible people': Negotiating 'queer community' on dating apps. *International Journal of Cultural Studies*, 24(3), 398–413. <https://doi.org/10.1177/1367877920959332>
- Race, K. (2015a). 'Party and play': Online hook-up devices and the emergence of PNP practices among gay men. *Sexualities*, 18(3), 253–275. <https://doi.org/10.1177/1363460714550913>
- Race, K. (2015b). Speculative pragmatism and intimate arrangements: Online hook-up devices in gay life. *Culture, Health & Sexuality*, 17(4), 496–511. <https://doi.org/10.1080/13691058.2014.930181>
- Ranzini, G., & Lutz, C. (2017). Love at first swipe? Explaining Tinder self-presentation and motives. *Mobile Media & Communication*, 5(1), 80–101. <https://doi.org/10.1177/2050157916664559>
- Ranzini, G., Rosenbaum, J. E., & Tybur, J. M. (2022). Assortative (online) dating: Insights into partner choice from an experimental dating app. *Computers in Human Behavior*, 127, 1–7. <https://doi.org/10.1016/j.chb.2021.107039>
- Renninger, B. J. (2019). Grindr killed the gay bar, and other attempts to blame social technologies for urban development: A democratic approach to popular technologies and queer sociality. *Journal of Homosexuality*, 66(12), 1736–1755. <https://doi.org/10.1080/00918369.2018.1514205>
- Rosser, B. R. S., West, W., & Weinmeyer, R. (2008). Are gay communities dying or just in transition? Results from an international consultation examining possible structural change in gay communities. *AIDS Care*, 20(5), 588–595. <https://doi.org/10.1080/09540120701867156>
- Rowe, M. S., & Dowsett, G. W. (2008). Sex, love, friendship, belonging and place: Is there a role for 'gay community' in HIV prevention today? *Culture, Health & Sexuality*, 10(4), 329–344. <https://doi.org/10.1080/13691050701843098>
- Sevi, B. (2019). The dark side of Tinder. *Journal of Individual Differences*, 40(4), 242–246. <https://doi.org/10.1027/1614-0001/a000297>
- Sevi, B., & Doğruyol, B. (2020). Looking from the bright side: The light triad predicts Tinder use for love. *Journal of Social and Personal Relationships*, 37(7), 2136–2144. <https://doi.org/10.1177/0265407520918942>
- Sharabi, L. L., & Timmermans, E. (2021). Why settle when there are plenty of fish in the sea? Rusbult's investment model applied to online dating. *New Media & Society*, 23(10), 2926–2946. <https://doi.org/10.1177/1461444820937660>
- Shield, A. D. (2019). *Immigrants on Grindr: Race, sexuality and belonging online*. Palgrave Macmillan. <https://doi.org/10.1007/978-3-030-30394-5>
- Stacey, L., & Forbes, T. D. (2021). Feeling like a fetish: Racialized feelings, fetishization, and the contours of sexual racism on gay dating apps. *The Journal of Sex Research*, 59(0), 372–384. <https://doi.org/10.1080/00224499.2021.1979455>
- Sumter, S. R., & Vandenbosch, L. (2019). Dating gone mobile: Demographic and personality-based correlates of using smartphone-based dating applications among emerging adults. *New Media & Society*, 21(3), 655–673. <https://doi.org/10.1177/1461444818804773>
- Sumter, S. R., Vandenbosch, L., & Ligtenberg, L. (2017). Love me Tinder: Untangling emerging adults' motivations for using the dating application Tinder. *Telematics and Informatics*, 34(1), 67–78. <https://doi.org/10.1016/j.tele.2016.04.009>
- Tewksbury, R. (2008). Finding erotic oases: Locating the sites of men's same-sex anonymous sexual encounters. *Journal of Homosexuality*, 55(1), 1–19. <https://doi.org/10.1080/00918360802129253>
- Thomas, M. F., Binder, A., & Matthes, J. (2022). The agony of partner choice: The effect of excessive partner availability on fear of being single, self-esteem, and partner choice overload. *Computers in Human Behavior*, 126, 1–10. <https://doi.org/10.1016/j.chb.2021.106977>
- Thompson, L. (2018). "I can be your Tinder nightmare": Harassment and misogyny in the online sexual marketplace. *Feminism & Psychology*, 28(1), 69–89. <https://doi.org/10.1177/0959353517720226>
- Timmermans, E., & De Caluwé, E. (2017a). Development and validation of the Tinder Motives Scale (TMS). *Computers in Human Behavior*, 70, 341–350. <https://doi.org/10.1016/j.chb.2017.01.028>
- Timmermans, E., & De Caluwé, E. (2017b). To Tinder or not to Tinder, that's the question: An individual differences perspective to Tinder use and motives. *Personality and Individual Differences*, 110, 74–79. <https://doi.org/10.1016/j.paid.2017.01.026>
- Timmermans, E., De Caluwé, E., & Alexopoulos, C. (2018). Why are you cheating on tinder? Exploring users' motives and (dark) personality traits. *Computers in Human Behavior*, 89, 129–139. <https://doi.org/10.1016/j.chb.2018.07.040>
- Timmermans, E., Hermans, A.-M., & Oprea, S. J. (2020). Gone with the wind: Exploring mobile daters' ghosting experiences. *Journal of Social and Personal Relationships*, 38(2), 783–801. <https://doi.org/10.1177/0265407520970287>
- Tinder. (2020, June 16). *Tinder*. <https://tinder.com>
- Tziallas, E. (2015). Gamified eroticism: Gay male "social networking" applications and self-pornography. *Sexuality & Culture*, 19(4), 759–775. <https://doi.org/10.1007/s12119-015-9288-z>

- Wang, S. (2020). Chinese affective platform economies: Dating, live streaming, and performative labor on Blued. *Media, Culture & Society*, 42(4), 502–520. <https://doi.org/10.1177/0163443719867283>
- Ward, J. (2017). What are you doing on Tinder? Impression management on a matchmaking mobile app. *Information, Communication & Society*, 20(11), 1644–1659. <https://doi.org/10.1080/1369118X.2016.1252412>
- Weiser, D. A., Niehuis, S., Flora, J., Punyanunt-Carter, N. M., Arias, V. S., & Hannah Baird, R. (2018). Swiping right: Sociosexuality, intentions to engage in infidelity, and infidelity experiences on Tinder. *Personality and Individual Differences*, 133, 29–33. <https://doi.org/10.1016/j.paid.2017.10.025>
- Wu, S. (2021). Domesticating dating apps: Non-single Chinese gay men's dating app use and negotiations of relational boundaries. *Media, Culture & Society*, 43(3), 515–531. <https://doi.org/10.1177/0163443720974240>
- Wu, S., & Trottier, D. (2021). Constructing sexual fields: Chinese gay men's dating practices among pluralized dating apps. *Social Media + Society*, 7(2), 1–14. <https://doi.org/10.1177/20563051211009014>
- Wu, S., & Ward, J. (2018). The mediation of gay men's lives: A review on gay dating app studies. *Sociology Compass*, 12(2), 1–10. <https://doi.org/10.1111/soc4.12560>
- Wu, S., & Ward, J. (2020). Looking for “interesting people”: Chinese gay men's exploration of relationship development on dating apps. *Mobile Media & Communication*, 8(3), 342–359. <https://doi.org/10.1177/2050157919888558>
- Xu, D., & Wu, F. (2019). Exploring the cosmopolitanism in China: Examining mosheng ren (“the stranger”) communication through Momo. *Critical Studies in Media Communication*, 36(2), 122–139. <https://doi.org/10.1080/15295036.2019.1566629>
- Yeo, T. E. D., & Fung, T. H. (2018). “Mr right now”: Temporality of relationship formation on gay mobile dating apps. *Mobile Media & Communication*, 6(1), 3–18. <https://doi.org/10.1177/2050157917718601>
- Zablotska, I. B., Holt, M., & Prestage, G. (2012). Changes in gay men's participation in gay community life: Implications for HIV surveillance and research. *AIDS and Behavior*, 16(3), 669–675. <https://doi.org/10.1007/s10461-011-9919-9>